Short necked Sauropod
A new "Amazing Dragon" dinosaur, officially known as Lingwulong shenqi has been discovered in China. **https://dgit.in/sep18-76-1**

Computing emotions
MIT engineers have developed a new ML model that allows computers to understand human emotions **https://dgit.in/sep18-76-2**

Light Bending

One of the fundamental truths of physics - the rectilinear motion of light - has certain exceptions

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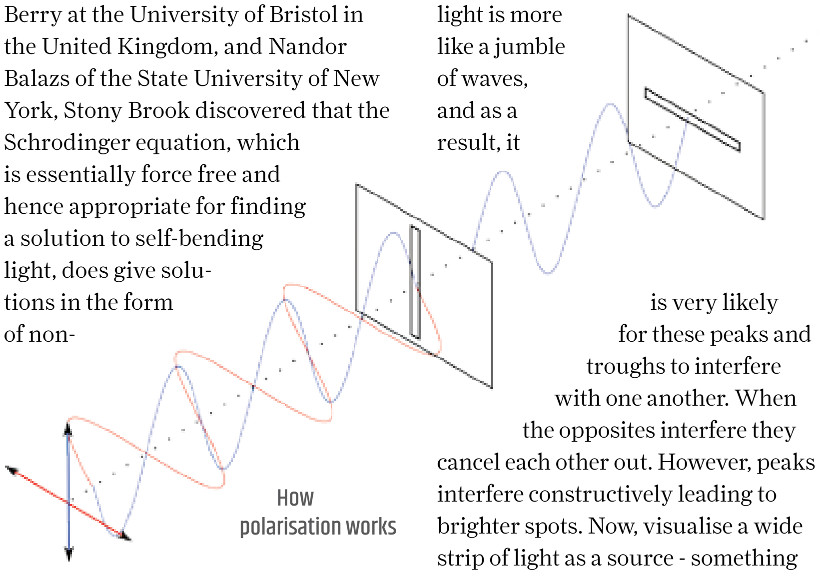
Irrespective of how far ahead we progress with new advancements in technology and science, the fundamental rules of this universe will still always govern us. The laws of physics, that have remained immutable for all of time will not suddenly change just because we've discovered something new, or understood something we didn't understand before, right? For instance, most us have been taught for ages that the motion of light is strictly rectilinear, i.e light always travels in a straight line. It in fact, does not, but that doesn't have to mean that the laws of physics themselves are flawed. It is our understanding of it that has been limited. That being said, there have been experiments that have demonstrated the bending nature of light - albeit under some very specific preconditions.

IT'S BEEN A WHILE

Light has always been able to bend. Remember high school physics? The phenomenon of refraction shows us that when light passes through media of varying density at an angle, its angle of propagation, as well as its speed, is modified. Which is why a spoon dipped into a glass of water appears to bend at its surface. Additionally, out in space, light bends around objects of massive gravitational pulls. In both cases, there's an external factor involved - the refractive index of media in the first, the gravitational pull of massive objects in the second. What we are looking for, however, is the demonstration of light bending by itself.

In the year 1979, physicists Michael Berry at the University of Bristol in the United Kingdom, and Nandor Balazs of the State University of New York, Stony Brook discovered that the Schrodinger equation, which is essentially force free and hence appropriate for finding a solution to self-bending light, does give solutions in the form of non-spreading "Airy" wave packets that freely accelerate even in the absence of any external potential. These are rays of light that, instead of diffracting over long distances, could bend, or rather accelerate sideways. Fast forward to 2007, Demetrios Christodoulides and other physicists at the University of Central Florida were able to generate optical versions of this Airy waveform. To do so, they had to manipulate laser light and the resultant waveform showed a slight curve as it passed through a detector. This was light that could bend on its own.



How polarisation works

To explain this, you have to remember that light is a wave, and like every wave, it comes with peaks and troughs. In actual application though, light is more like a jumble of waves, and as a result, it is very likely for these peaks and troughs to interfere with one another. When the opposites interfere they cancel each other out. However, peaks interfere constructively leading to brighter spots. Now, visualise a wide strip of light as a source - something like an incandescent tube but as a laser. Now, by controlling the initial points of the emerging waveforms along the strip, it is possible to make the waves interfere constructively in a way that they appear to form a curve and cancel out at all points not on the curve. This effectively is an optical illusion and can only bend light upto 8 degrees.

However, in 2012, a group of physicists at Israel Institute of Technology,